



A replica of George Cayley's glider, which made the world's first fixed-wing flight in 1853 in Yorkshire, England.

“ I AM WELL CONVINCED THAT **AERIAL NAVIGATION WILL FORM A MOST PROMINENT FEATURE IN THE PROGRESS OF CIVILIZATION.** ”

George Cayley, British aeronautical engineer, 1804

GREAT AUKS HAD BEEN EASY TARGETS FOR HUNTERS in areas around the North Atlantic for many years, and by the 1840s they were practically extinct. **The last bird was spotted** off the coast of Newfoundland in Canada in 1852.

British physicists **James Joule** (1818–89) and **William Thomson** (1824–1907) discovered the Joule–Thomson effect in 1852. This explains **the way that gases and liquids cool and expand** after flowing through a restriction or throttle. The Joule–Thompson effect is central to the way refrigerators and air conditioning systems work.

The age of aviation began on September 24, 1852 when French engineer **Henri Giffard** (1825–82) made the **first powered and controlled flight**, flying 17 miles (27 km) from Paris to Trappes in France. The flight was made in a powered airship, which consisted of a cigar-shaped, hydrogen-filled balloon that provided the lift, and a steam-driven propeller that moved it through the air.

In 1853 there was another aviation first, the **first flight in a full-sized aircraft**—a glider built by the British engineer **George Cayley** (1773–1857), who had pioneered understanding of the theory of flight. Details of the flight, across Brompton Dale in Yorkshire, England, are not clear; some reports say Cayley's butler was

Great Auk
The last Great Auk, one of the largest birds of the North Atlantic at 31.5 in (80 cm) tall, was last seen in 1853.

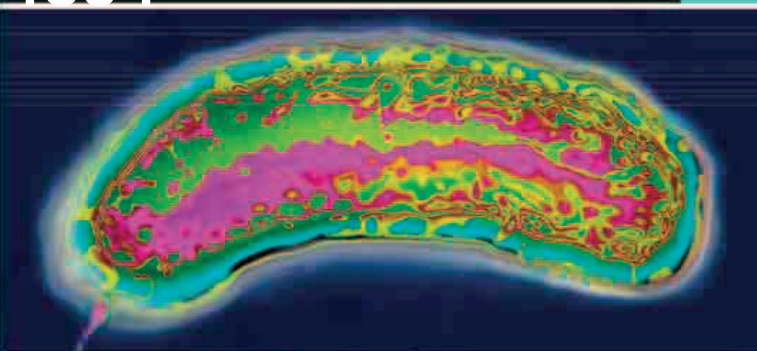


the pilot, while others say it was his footman. Nonetheless, it was a historic achievement.

1853 was also an important year for medicine and physiology. French physiologist **Claude Bernard** (1813–78) discovered that **glucose sugar, the body's energy food, is stored temporarily in the liver** in the form of a starchlike substance called glycogen, ready to be released into the blood as glucose when energy is needed.

French physician **Antoine Desormeaux** (1815–82) **developed the endoscope** for surgical operations. This was a long metal tube that could be inserted into the body to make examinations, using light from a paraffin-fueled lamp reflected in a mirror.

Another Frenchman, surgeon **Charles Pravaz** (1791–1853), and British physician **Alexander Wood** (1817–84) independently **invented a practical hypodermic syringe** with a hollow metal needle that could be inserted into the body to deliver drugs directly into veins for much faster effect than taking the drugs by mouth.



This color-enhanced micrograph shows the cholera bacterium, now known to be responsible for cholera.

FOLLOWING A SUGGESTION

BY BRITISH PHYSICIST William Thomson, German physicist **Hermann von Helmholtz** (1821–94) and British engineer **William Rankine** (1820–72) developed Rudolf Clausius's theory that heat never flows from a cold to a hot place. This implies that heat will ultimately spread evenly through the Universe and once this happens energy will not be able to move and the Universe will come to a stop in what is called “**heat death of the Universe.**”

British astronomer **George Airy** (1801–92) calculated the **density of Earth** by measuring the swing of a pendulum on Earth's surface and 1,257 ft (383 m) down a coal mine. Different measurements revealed slight variations in the effects of gravity, from which he obtained the figure 3.795 oz/in³ (6.566 g/cm³). Today's accepted figure is 3.19 oz/in³ (5.52 g/cm³).

Two new types of mathematics were introduced in 1854. One was the non-Euclidean geometry of German mathematician **Bernhard Riemann** (1826–66). Euclidean geometry applies only to flat surfaces; **Riemannian geometry is the geometry of curved surfaces**, important because the surface of Earth is curved. In Euclidean geometry, the angles in a triangle add up to two right angles and the shortest

distance between two points is a straight line, in Riemannian geometry angles in a triangle add up to more than two right angles and there is no such thing as a straight line along a surface.

The new algebra of British mathematician **George Boole** (1815–64) was intended to make logic mathematical, not

3.19
OZ/IN³
THE AVERAGE DENSITY OF EARTH

philosophical. Boole argued that any proposition could be reduced to just “and,” “or,” and “not,” and worked through to a conclusion. Today, **Boolean logic** combines with the binary system of numbers to shape all computer programs.

In August, there was an outbreak of cholera in London's Soho. British physician **John Snow** (1813–58) traced the source to a single water pump, thus validating his theory that **cholera is water-borne.**

1852 The last Great Auk seen off Newfoundland, Canada

1852 James Joule and William Thomson discover the Joule–Thomson effect

September 24, 1852 Henri Giffard's airship makes the first powered controlled flight

1853 Claude Bernard realizes that glucose is stored in the liver as glycogen

1853 George Cayley's glider makes the first manned flight in a full-sized aircraft

1853 Antoine Desormeaux creates the endoscope

1853 Charles Pravaz and Alexander Wood invent the hypodermic syringe

Hermann von Helmholtz and William Rankine theorize about the heat death of the Universe

August 31–September 1 John Snow traces the source of a cholera outbreak in London

George Airy measures the density of Earth

George Boole introduces Boolean logic and algebra

Bernhard Riemann introduces Riemannian geometry